

Lily (Shi-tao) Zhang

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PROFESSIONAL SUMMARY

Healthcare data analytics and research professional with a Ph.D in Biochemistry and Molecular Biology and a Master's degree in Epidemiology and Biostatistics. Combines extensive biomedical research experience with advanced training in statistical analysis, study design, and healthcare data analytics. Skilled in R, SAS, and Python for data wrangling, statistical modeling, survival analysis, data visualization, and interpretation of complex clinical and biomedical datasets. Experienced in translating quantitative findings into clear reports, presentations, and actionable insights through interdisciplinary collaboration.

EDUCATION

Case Western Reserve University, Cleveland, OH

Master's in Epidemiology and Biostatistics, Health Analytics Track

Aug 2025 – Aug 2026

Jilin University, Changchun, China

Ph.D in Biochemistry and Molecular Biology

Sep 2015 – Dec 2021

Jilin University, Changchun, China

Bachelor's in Life Science

Sep 2011 – Dec 2015

DATA ANALYTICS PROJECTS

Gut Microbiome Composition and Differential Abundance Across Dietary Groups

Analyzed 16S rRNA sequencing data to evaluate differences in gut microbial diversity, community composition, and differential abundance between long-term dietary groups.

Tools & Methods: R, DADA2, phyloseq, RDP taxonomy, alpha diversity, Bray-Curtis dissimilarity, PCoA, PERMANOVA, dispersion testing, ANCOM-BC2, FDR correction

Inflammatory Biomarkers and Frailty Risk in People Aging with HIV

Analyzed clinical and biomarker data to investigate associations between inflammatory extracellular-vesicle markers, frailty status, and longitudinal health outcomes among people aging with HIV.

Tools & Methods: R, data cleaning and transformation, Wilcoxon test, multivariable regression, Cox proportional hazards models, proportional hazards diagnostics

Healthcare Utilization During Florida Hurricanes using HCUP data

Analyzed HCUP Florida administrative healthcare data to evaluate patterns and changes in healthcare utilization associated with hurricane-related periods.

Tools & Methods: R, SAS, HCUP administrative healthcare data, data management, descriptive statistics, regression analysis

Health and Chronic Disease Among Older Adults Using RAND Health and Retirement Study (HRS)

Analyzed RAND HRS data to investigate health and chronic disease patterns among older adults and assess associations with demographic and health-related characteristics.

Tools & Methods: R, RAND HRS, survey data management, descriptive statistics, regression analysis, data visualization

RESEARCH EXPERIENCE

Conformational Analysis of Human Estrogen Receptor α Bound to Estrogen Response Elements

Investigated the structural and biochemical mechanisms underlying human estrogen receptor α (hER α) binding to estrogen response elements (ERE) through protein characterization and structural analysis.

Tools & Methods: Protein expression and purification, FPLC, EMSA, single-particle cryo-EM, structural analysis, PyMOL

Molecular Dynamics Analysis of Type 2 Diabetes-Related Proteins and Ligand Interactions

Investigated molecular interactions, conformational dynamics, and potential binding mechanisms of proteins associated with type 2 diabetes using computational molecular modeling and molecular dynamics simulations.

Tools & Methods: AMBER, Linux, R, Python, molecular dynamics simulation, molecular docking, trajectory analysis, PyMOL, VMD

TECHNICAL SKILLS

Statistical Programming: R, SAS, Python

Data Analysis: Data cleaning, exploratory data analysis, linear regression, generalized linear models, logistic regression, survival analysis, hypothesis testing, multiple-testing correction, principal component analysis, machine learning fundamentals

Healthcare / Research Analytics: Study design, clinical and biomedical data analysis, biomarker analysis, reproducible analytical workflows, data visualization, scientific reporting

R Ecosystem: tidyverse, ggplot2, phyloseq, DADA2, survival, rstanarm, tidymodels, gt, janitor, patchwork

Other Tools: Excel, PowerPoint, Word, Linux, MATLAB, PyMOL, VMD

Laboratory / Structural Biology: Protein expression and purification, FPLC, Western blotting, EMSA, cell culture, single-particle cryo-electron microscopy

PUBLICATIONS

Peer-Reviewed Publications

1. **Zhang S**, Zuo C, Li W, Fu X, Xing S, Zhang X. Identification of key genes associated with the effect of estrogen on ovarian cancer using microarray analysis. *Archives of Gynecology and Obstetrics*. 2016;293(2):421–427. doi:10.1007/s00404-015-3833-8.
2. **Zhang S**, Zhang X, Fu X, Li W, Xing S, Yang Y. Identification of common differentially-expressed miRNAs in ovarian cancer cells and their exosomes compared with normal ovarian surface epithelial cells. *Oncology Letters*. 2018;16(2):2391–2401. doi:10.3892/ol.2018.8954.
3. **Zhang S**, Wang Y, Han L, Fu X, Wang S, Li W, Han W. Targeting N-Terminal Human Maltase-Glucoamylase to Unravel Possible Inhibitors Using Molecular Docking, Molecular Dynamics Simulations, and Adaptive Steered Molecular Dynamics Simulations. *Frontiers in Chemistry*. 2021;9:711242. doi:10.3389/fchem.2021.711242.
4. **Zhang S**, Lv S, Fu X, Han L, Han W, Li W. Molecular Dynamics Simulations Study of the Interactions between Human Dipeptidyl-Peptidase III and Two Substrates. *Molecules*. 2021;26(21):6492. doi:10.3390/molecules26216492.

Manuscripts Under Review / In Preparation

5. Zhang Z, **Zhang S**, Gregor W, Yu E, Yang S. Three DNA conformations at a single ER -ERE interface. *Manuscript under review*.
6. Zhang Z, **Zhang S**, Yu E, Yang S. ER -ERE binding conformations. *Manuscript in preparation*.
7. **Zhang S**, Wilson B, Freeman M. *Manuscript in preparation*.